Fri 28 Mar 2025

Last time we saw that another approach for finding x^* such that $f(x^*)=0$ is to instead find <u>fixed points</u> $x=p^*$ of $y=g(x)$ (that is, p^* such that $g(p^*)=p^*$) where $g(x)$ is chosen so that fixed points of $g(x)$ are roots of $f(x)$.

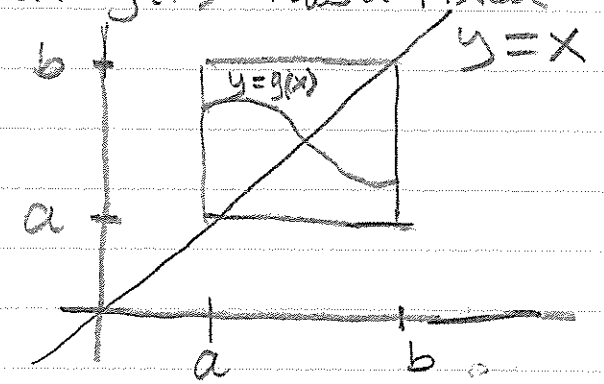
- We proposed finding fixed points by using the iteration $x_{i+1}=g(x_i)$, x_0 given. Some questions that you might have are:

  - how can we know that $g(x)$ has a fixed point?
  - ——— " ——— a fixed point is unique?
  - how can we know when the fixed point iteration will converge.
  - can we know that it won't converge?

Fortunately, we have some theorems that can help us answer those questions.

<u>Theorem</u> (a) If $g \in C[a,b]$ with $a>b$ and $\forall x \in [a,b]$, $g(x) \in [a,b]$, then $g(x)$ has a fixed point in $[a,b]$

picture:



and (b) If, in addition, $g'(x)$ exists on (a,b) and $|g'(x)| \le k < 1$ for some constant k and all $x \in (a,b)$, then the fixed point is unique.

This is not too difficult to prove.

Proof: (a) If $g(a) = a$ or $g(b) = b$, then
(existence) $g(x)$ has a fixed point in $[a,b]$.

Otherwise, if $g(a) \neq a$ and $g(b) \neq b$,

then $g(a) > a$ and $g(b) < b$,
since $g(x) \in [a,b]$ $\forall x \in [a,b]$.

so $g(a) - a > 0$ and $g(b) - b < 0$

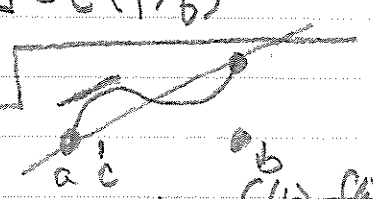
Let $h(x) = g(x) - x$. Then
1. $h(x)$ is continuous on $[a,b]$
2. $h(a) > 0$
3. $h(b) < 0$

$\therefore$ by the Intermediate Value Theorem, $h(x)$ has a root between a and b. i.e., $\exists\, x^* \in (a,b)$ such that $h(x^*) = 0$

$\Longrightarrow$ $g(x^*) - x^* = 0$ or $g(x^*) = x^*$ and x^* is a fixed point.

(b) (uniqueness) Suppose p, q are two fixed points of $g(x)$ in $[a,b]$ and $p \neq q$.

Then by the Mean Value Theorem, $\exists\, c \in (p,q)$ such that $g'(c) = \dfrac{g(p) - g(q)}{p - q}$



MVT: If $f \in C[a,b]$ and f is differentiable on (a,b) then a number $c \in (a,b)$ exists with $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

Since p, q are fixed points of $g(x)$

Then we have

$$|p-q| = |g(p) - g(q)|$$
$$= |g'(c)|\,|p-q| \quad \text{(by MVT)}$$
$$\le k\,|p-q|$$
$$< |p-q|$$

But $|p-q| \nless |p-q|$, so we have a contradiction and $\therefore$ $p=q$, the fixed point is unique.

example: $g(x) = 0.2\sin(x) + 0.5$ and interval $[-1,1]$

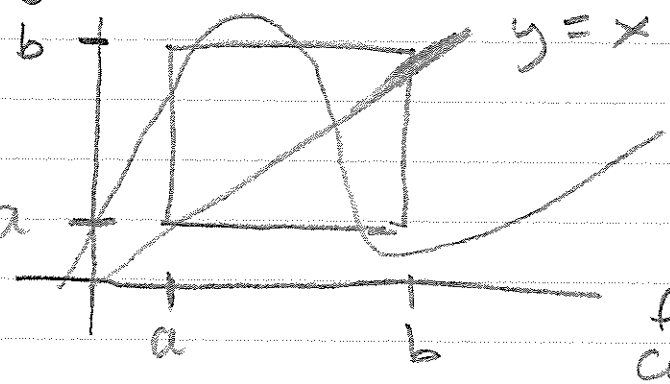
Since $\sin(x) \in [-1,1]$, $0.2\sin(x) \in [-0.2, 0.2]$ and $g(x) \in [0.3, 0.7] \subset [-1,1]$.

$\therefore$ there is a fixed point of $g(x)$ in $[-1,1]$.

Also, $g'(x) = 0.2\cos(x)$, so $|g'(x)| \le 0.2 < 1$ for $x \in [-1,1]$, and so the fixed point is unique.

<u>Note</u>: The conditions in the statement of the Theorem are sufficient conditions for a fixed point but they are not necessary. (They need not be satisfied)

picture



There is a fixed point but conditions <u>not</u> met.

A related question:
When will the fixed point iteration converge?

It turns out that the question of whether a fixed point iteration converges to a fixed point x^* is related to the size of $g'(x^*)$.

Suppose $g(x^*) = x^*$

and suppose that at some point in the fixed point iteration got you to the point $x_i = x^* + h$, where h is small.

Then
$$\begin{aligned} x_{i+1} &= g(x_i) \\ &= g(x^* + h) \\ &= g(x^*) + h g'(x^*) + \frac{h^2}{2!} g''(x^*) + \dots \\ &\doteq g(x^*) + h g'(x^*) \end{aligned}$$

so $x_{i+1} - g(x^*) \doteq h g'(x^*)$

or $x_{i+1} - x^* \doteq h g'(x^*)$

We know $|x_i - x^*| = |h|$

so
$$\begin{aligned} \underbrace{|x_{i+1} - x^*|}_{\text{error in new approx}} &\doteq |g'(x^*)|\,|h| \\ &= |g'(x^*)|\,\underbrace{|x_i - x^*|}_{\text{error in current approx.}} \end{aligned}$$

if $|g'(x^*)| < 1$, x_{i+1} is closer to x^* than x_i

but if $|g'(x^*)| > 1$, x_{i+1} is further from x^* than x_i

$|g'(x^*)| < 1$: x^* is called an attracting fixed point and the fixed point iteration will converge to the fixed point if we get close enough

$|g'(x^*)| > 1$: x^* is called a repelling fixed point and the fixed point iteration won't converge to it.

<u>example</u>: ① $g(x) = \frac{1}{2}(x^3 + x)$

What are it's fixed points? $x=0$ $x=1$ $x=-1$

$g'(x) = \frac{1}{2}(3x^2+1)$ $g'(0) = 1/2$ $g'(1) = 2$ $g'(-1) = 2$

We can sketch the trajectories of various iterations (observe: divergence, convergence, divergence) (see next page)
$x = \pm 1$ are unstable fixed pts. $x=0$ is a stable fixed point

② Now consider $g(x) = -\frac{1}{2}(x^3 + x)$ (negation of ①)

There is one fixed point: $x=0$

$g'(x) = -1/2(3x^2+1)$ $g'(0) = -1/2$

iteration sketch of trajectories shows: convergence

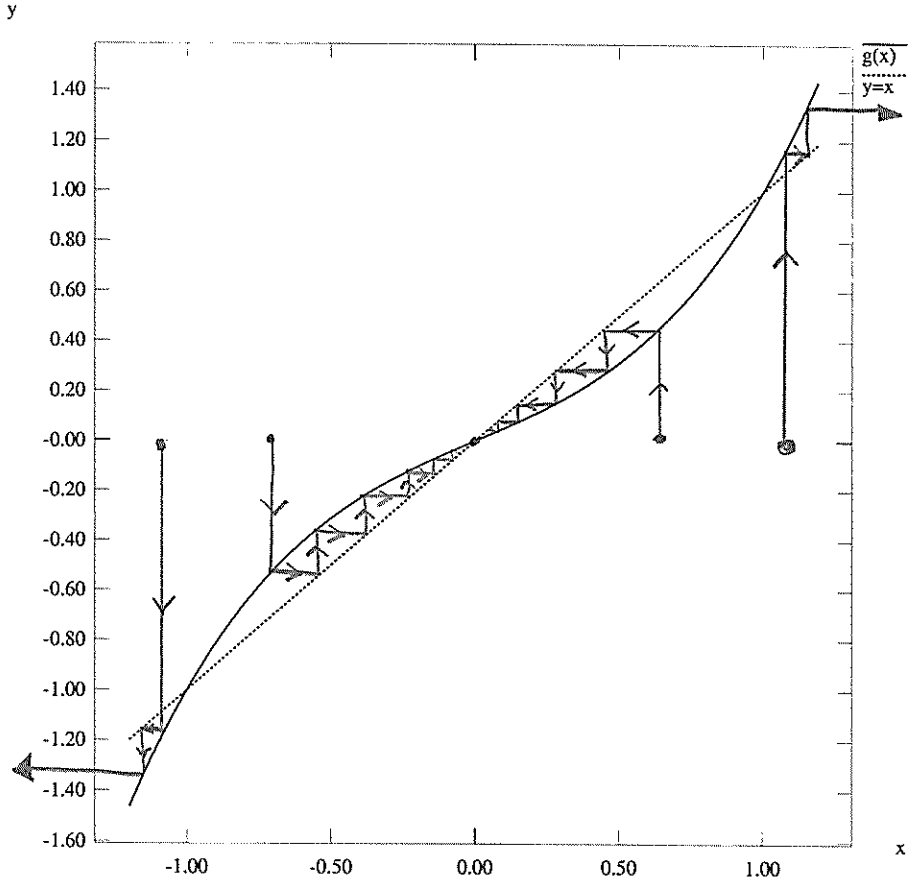
period 2 points

$(g(1) = -1,\ g(-1) = 1)$

divergence

We can prove a theorem about guaranteeing convergence.

y = g(x) = 0.5 (x3 + x) and y = x**

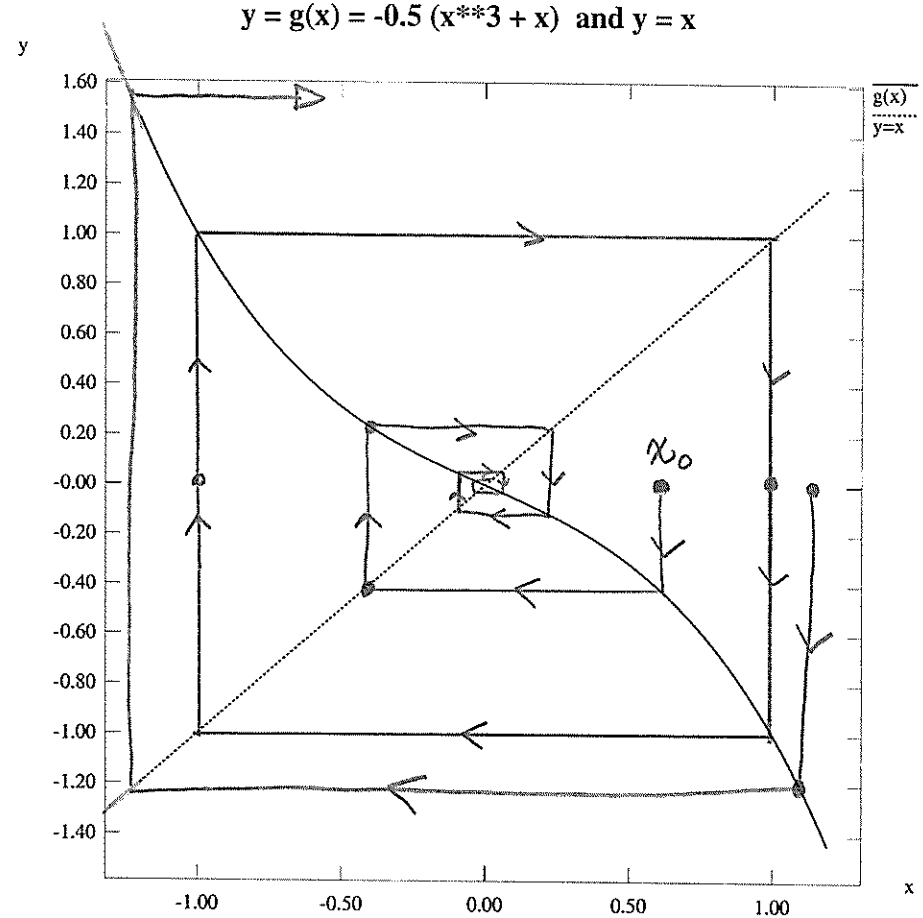


$g(0) = 0$ $\quad g'(0) = 0.5$

$g(1) = 1$ $\quad g'(1) = 2$

$g(-1) = -1$ $\quad g'(-1) = 2$

y = g(x) = -0.5 (x3 + x) and y = x**



$g(0) = 0$ $\quad g'(0) = -0.5$ $\quad g'(1) = -1$

$g'(-1) = -1$

$g(1) = -1$

$g(-1) = 1$ $\quad \pm 1$ are period-2 points

$g' = -\frac{1}{2}(3x^2+1)$

basin of attraction for $x^* = 0$ is $-1 < x_0 < 1$